

Returns and volume: Frequency connectedness in cryptocurrency markets[☆]Panos Fousekis^{*}, Dimitra Tzaferi

Aristotle University, Greece



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ABSTRACT

The present paper investigates the causal link between returns and volume in cryptocurrency markets. Relative to earlier empirical studies on the topic, it employs a flexible methodological approach that allows the intensity and the pattern of the link to vary with the traders' planning horizon (shorter vs longer runs) and the nature of the preceding price shocks (positive vs negative). Causality is bi-directional implying that: (a) volume-based technical analysis can lead to higher profits and (b) trend-following is likely to be a common investment strategy among rational uninformed traders. There is, therefore, scope for establishing appropriate disclosure systems to deal with information asymmetry and the risk associated with it. In the longer-run, spillovers from returns to volume are stronger than those in the opposite direction. The same, generally, applies to spillovers from positive relative to negative returns. Consequently, returns (especially the positive ones) are more likely to induce fundamental changes in traders' expectations compared to volume.

1. Introduction

Bitcoin, the first open-source digital currency, was introduced in 2008 by a person or a group of persons under the pseudonym Satoshi Nakamoto. It was based on a shared and encrypted database technology called blockchain. The most striking feature of bitcoin and the associated with it payment system was the absence of a governing and controlling central authority; bitcoin's supply was limited by the design of a protocol and parties were allowed to make transactions without the intervention of a trusted third party (e.g. Nakamoto, 2008; Yermack, 2014; Bouri et al., 2017; Bouri et al., 2018). Over the last ten years, a very large number of alternative cryptocurrencies (altcoins) have been developed and released mainly through imitation. According to coinmarketcap (<https://coinmarketcap.com>) in March 2020, there were more than 5000 cryptocurrencies with a total market capitalization of 185 billion USD.

The growth in cryptocurrency markets has attracted the attention of investors, media, and regulatory authorities. At the same time, it has been accompanied by a flourishing academic interest from very diverse fields such as computer science, law, economics, and finance. The first issue research economists examined was whether cryptocurrencies fulfil the three functions of any bona fide currency (i.e. medium of exchange, unit of account, and store of value). Given that their level of acceptance is still

low, their price variability is very high, and their unit values are tradable, a consensus has emerged early in the relevant literature that cryptocurrencies are a new category of investment assets than just a new form of money (e.g. Glaser et al., 2014; Yermack, 2014; European Central Bank, 2015; Polasik et al., 2015). That, in turn, has spurred new research efforts towards assessing the role of cryptocurrencies in financial markets and portfolio management.

Baek and Elbeck (2015) and Cheung et al. (2015) provided evidence that Bitcoin returns were not affected by fundamental economic factors, while Cheah and Fry (2015) identified a bubble in the Bitcoin market; they concluded that Bitcoin had a mostly speculative value. Further evidence of Bitcoin bubble behaviour was offered by the recent studies of Chaim and Laurini (2019) and Geuder et al. (2019). Li and Wang (2017), however, showed that Bitcoin price is sensitive to macro-economic indicators. Kristoufek (2015) and Selgin (2015) argued that Bitcoin is a unique asset combining features from commodities (precious metals, in particular) and sovereign currency and, thus, it can be viewed as synthetic commodity money.¹

Several empirical studies have focused on the interdependence between cryptocurrencies and traditional assets (stocks, bonds, hard currencies, commodities, and real estate) in an attempt to verify whether cryptocurrencies may serve as diversifying, hedge, or safe-haven

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^{*} Corresponding author.

E-mail address: fousekis@econ.auth.gr (P. Fousekis).

¹ Under the Commodity Exchange Act (CEA) virtual currencies have been determined to be commodities (https://www.cftc.gov/sites/default/files/idc/groups/public/%40customerprotection/documents/file/oceo_bitcoinbasics0218.pdf).

investments. Briere et al. (2015) and Symitsi and Chalvatziis (2018) found that Bitcoin had a low correlation with traditional assets and its inclusion in a diversified portfolio led to a substantial risk reduction. In the paper of Bouri et al. (2017), Bitcoin turned out to be an effective diversifier against movements in a large number of different asset classes and, in a few cases, even a safe-haven one. Very similar were the results of the study by Urom et al. (2020). Bouoiyour et al. (2019) also reported that Bitcoin might serve as a safe-haven against US stock losses (but only in the short-run).

A number of studies have examined the price formation (returns and volatility) in cryptocurrency markets. Kristoufek (2013) and Bouoiyour and Selmi (2015) found a strong bi-directional causality between Bitcoin prices and search queries for Bitcoin in the Web. Balcilar et al. (2017) and Bouri et al. (2019a) sought evidence regarding price formation by focusing on the micro-structure of cryptocurrency markets.² In particular, Balcilar et al. (2017) employed a causality-in-quantiles test to investigate spillovers from volume to returns and to volatility for the Bitcoin market. According to their results, volume did not Granger-cause volatility but it had some predictability over returns around the median. Bouri et al. (2019a) performed a similar investigation for Bitcoin and six altcoins using the copula-quantile causality approach. They reported that volume caused extreme positive and extreme negative returns for all cryptocurrencies considered, but it could predict volatility only for Litecoin, NEM, and Dash. Naeem et al. (2020) explored the relationship between returns and volume for Bitcoin, Ethereum, and Litecoin using Garch-copula models. Their results pointed to asymmetric contemporaneous tail dependence (i.e. tail dependence turned out to be stronger when returns and volume were high than when returns and volume were low). Alaoui et al. (2019), using a multi-fractal correlation approach, found that the mechanism governing the return-volume dynamics for Bitcoin is characterized by non-linear dependency and multifractality; as a consequence, traders' inferences about Bitcoin price cannot be complete without simultaneous inferences on Bitcoin trading volume (and vice versa).

The issue of whether there exists a causal order from volume to returns is important as it is related directly to the informational efficiency of a market. Evidence that volume has an information content about future price changes is against the efficient market hypothesis (EMH) (Fama, 1970) and it may explain why traders who include volume measures in their technical analysis tend to perform better than those who do not (e.g. Blume et al., 1994; Dormeier, 2011).³ Moreover, an asymmetric response of price to volume has been employed in several earlier theoretical and empirical studies to explain market crashes (e.g. Gennotte and Leland, 1990; Marsh and Wagner, 2004). As noted by Griffin et al. (2007) and Chen (2012), however, the question of whether investors trade more when a market has done well in the past is also relevant and interesting; if this is indeed the case, market makers and liquidity providers may forecast trading intensity and portfolio managers may devise efficient trading strategies. Neither Balcilar et al. (2017) nor Bouri et al. (2019a) explored the possibility of causal order from returns to volume.⁴ This, despite the existence of plenty of empirical evidence from traditional asset markets that causality may also flow from price changes to market activity (e.g. Hiemstra and Jones, 1994; Fujiwara and Mougue, 1997; Griffin et al., 2007; Chuang et al., 2009; Gebka and

Wohar, 2013; He et al., 2014; Kumar, 2017).

Against this background, the objective of the present paper is to shed more light on the lag-lead relationship between returns and trading activity in cryptocurrency markets. This is pursued using frequency connectedness analysis, an econometric tool proposed by Barunik and Krehlik (2018), and data from four major cryptocurrencies over 2018 to 2020. Frequency connectedness analysis allows a shock in a given stochastic process to impact on another stochastic process at different frequencies (e.g. short-, medium-, and long-run) with different strengths. As such, it is highly suitable to accounting for the possibility that agents (here traders in cryptocurrency markets) operate on different investment horizons; moreover, it can be naturally seen as causality in the time-frequency domain. Barunik and Krehlik (2018), Barunik and Kocenda (2019), and Tiwari et al. (2020) employed frequency connectedness to study risk (volatility) spillovers among major financial firms, oil and FOREX markets in the US, and energy, food, industry, agricultural, and metal markets, respectively; the same approach has been also employed by Gillaizeau et al. (2019) to quantify volatility spillovers in cross-market Bitcoin prices (Bitcoin/USD, Bitcoin/EURO, Bitcoin/AUD, Bitcoin/CAD, and Bitcoin/GBP).

To the best of our knowledge, there have been no earlier empirical studies on the micro-structure of cryptocurrency markets using frequency connectedness analysis. This paper places special emphasis on the identification and quantification of different types of asymmetry. For the temporal returns-volume link, asymmetry may be manifested in the relative strength of spillovers: (a) from returns to volume and from volume to returns; (b) to volume or to returns arising from positive and from negative price shocks⁵; and (c) to volume or to returns at different frequencies. In what follows, section 2 presents the analytical framework, and section 3 the data, the empirical models and the results. Section 4 offers conclusions and suggestions for future research.

2. Frequency, within, and asymmetric connectedness

To develop connectedness (spillovers) measures, Diebold and Yilmaz (2012) relied on the forecast error variance decomposition (FEVD) matrix of a generalized vector autoregressive (VAR) model. Let $X_t = (X_{1,t}, \dots, X_{N,t})$ (with $t = 1, \dots, T$) a N -dimensional stationary stochastic process and let also the associated with it VAR(p) model $X_t = \sum_{k=1}^p \Phi_k X_{t-k} + \varepsilon_t$ (where Φ_k denote coefficient matrices and $\varepsilon_t \sim N(0, \Sigma_\varepsilon)$ a vector of iid disturbances). Given that the VAR(p) is invertible, it has a moving average representation with an infinite order ($MA(\infty)$) $X_t = \sum_{h=0}^{\infty} B_h \varepsilon_{t-h}$, where the $N \times N$ coefficient matrices B_h are obtained as $B_0 = I_N$ and $B_h = \sum_{j=1}^p \Phi_j B_{h-j}$. The ij th element of the $N \times N$ FEVD matrix at the H -step-ahead forecast horizon is computed as

$$\theta_{ij}^H = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' B_h \Sigma_\varepsilon e_j)^2}{\sum_{h=0}^{H-1} (e_i' B_h \Sigma_\varepsilon B_h' e_i)}, \quad (1)$$

where $ij = 1, \dots, N$ and $e_i(e_j)$ are selection vectors with 1 at the i th (j th) element and 0 elsewhere. θ_{ij}^H traces how much of the future forecast error variance of the variable i is due to innovations in variable j . The normalized by the row sum elements of the FEVD matrix ($\tilde{\theta}_{ij}^H = \theta_{ij}^H / \sum_{j=1}^N \theta_{ij}^H$) are the building blocks of four alternative measures of connectivity; namely, *total*, *directional*, *net*, and *net pairwise*.

² The idea behind micro-structure modelling is that, in the short- and the medium-run, price dynamics are more likely to be influenced by micro-variables (such as the market activity/trading volume) than by market fundamentals (e.g. Kumar, 2017; Fousekis and Tzaferi, 2020).

³ Urquhart (2016), Wei (2018), and Bouri et al. (2019b) assessed the informational efficiency of cryptocurrencies markets using alternative random walk and long-memory tests. They concluded that past returns can be employed to predict future ones. According to the results by Wei (2018), the informational inefficiency tended to be inversely related to market liquidity.

⁴ Alaoui et al. (2019) emphasized that the presence of non-linearity and multifractality cannot be interpreted as causality.

⁵ Jalal et al. (2020) employed quantile regressions to investigate herding behaviour in cryptocurrency markets. Their results indicated that, in bullish and high volatility periods, overexcitement may lead to high trading activity. Baur and Dimpfl (2018), using the TGARCH model, found that positive price shocks in cryptocurrency markets tend to increase price volatility more than the negative ones.

Total connectedness (S^H) is the sum of the off-diagonal elements to the sum of all elements of the normalized FEVD matrix,

$$S^H = 100 \frac{\sum_{i,j=1, i \neq j}^N \tilde{\theta}_{i,j}^H}{\sum_{i,j=1}^N \tilde{\theta}_{i,j}^H} = 100 \frac{\sum_{i,j=1, i \neq j}^N \tilde{\theta}_{i,j}^H}{N} \quad (2)$$

It gives the share of the total forecast variance contributed by all stochastic cross-shocks to the elements of X_t . Directional measures provide information on the connectivity between the overall system and individual elements of it. In particular, the directional index of shocks received by element i from the remaining $N-1$ elements is computed as

$$S_{i \leftarrow \bullet}^H = 100 \frac{\sum_{j=1, j \neq i}^N \tilde{\theta}_{j,i}^H}{N} \quad (3)$$

whereas its counterpart of shocks transmitted by element i to the rest is computed as

$$S_{i \rightarrow \bullet}^H = 100 \frac{\sum_{j=1, j \neq i}^N \tilde{\theta}_{i,j}^H}{N} \quad (4)$$

The net spillover measure for element i is the difference

$$S_i^H = S_{i \rightarrow \bullet}^H - S_{i \leftarrow \bullet}^H \quad (5)$$

and it provides information on whether the element in question is a net contributor or a net receiver of shocks. Finally, the net pairwise spillover focuses on the connectedness between just two stochastic processes and it is computed as

$$S_{i,j}(H) = 100 \frac{\tilde{\theta}_{j,i}^H - \tilde{\theta}_{i,j}^H}{N} \quad (6)$$

When a VAR is 2-dimensional (as in this paper) causality in any of the two directions is assessed by the magnitude and the statistical significance of the $S_{i \rightarrow \bullet}^H$ ($i = 1, 2$) measures. Also, the net pairwise spillover coincides with the net spillover.

Frequency connectedness measures the proportion of the FEVD, at a given frequency band (ω), explained by shocks in one or more variables. For its measurement, Fourier transforms of impulse response functions (i.e. frequency responses) are utilized to define the spectral representation of the FEVD (Barunik and Krehlik, 2018). The response at ω is $B(e^{-i\omega}) = \sum_h e^{-i\omega h} B_h$ and the power spectrum of X_t (that is, the Fourier transform of the process $MA(\infty)$) is

$$S_X(\omega) = \sum_{h=-\infty}^{\infty} E(X_t X'_{t-h}) e^{-i\omega h} = B(e^{-i\omega}) \Sigma B'(e^{i\omega}). \quad (7)$$

The spectral density in (7) describes how the variance of the N -dimensional stochastic process is distributed over the frequency components ω . Using the spectral representation for covariance $E(X_t X'_{t-h}) = \int_{-\pi}^{\pi} S_X(\omega) e^{i\omega h} d\omega$, Barunik and Krehlik (2018) defined the variance decomposition on the frequency band $d = (\alpha, \beta) : \alpha, \beta \in (-\pi, \pi), \alpha < \beta$ as

$$\theta_{ij}(d) = \frac{1}{2\pi} \int_{\alpha}^{\beta} \Gamma_i(\omega) (\psi(\omega))_{ij} d\omega, \quad (8)$$

where $(\psi(\omega))_{ij}$ is the causation spectrum

$$\psi(\omega)_{ij} = \frac{\sigma_{ij}^{-1} (e' B(e^{-i\omega}) \Sigma e_i)^2}{(B(e^{-i\omega}) \Sigma B'(e^{-i\omega}))_{ij}} \quad (9)$$

and $\Gamma_i(\omega)$ is a weighting function. $\psi(\omega)_{ij}$ stands for the portion of the spectrum of the i th variable at frequency ω due to shocks in the j th variable. Barunik and Krehlik (2018) interpreted it as within frequency causation. $\Gamma_i(\omega)$ is the share (power) of the spectral density at ω . Barunik and Krehlik (2018) termed $\theta_{ij}(d)$ as frequency connectedness. Relative to within frequency causation, frequency connectedness appears to be a more complete measure in the sense it is likely to be closer to the total connectedness between two time series. The values of $\theta_{ij}(d)$ (and/or of $\psi(\omega)_{ij}$), after being normalized by the respective row sums, can be substituted into equations (2)–(7) to yield alternative measures of within causation and connectedness at the frequency domain.

Asymmetric frequency connectedness (or within frequency causation) may occur both at a given frequency as well as across frequencies. A potential source of asymmetry at a given frequency may be the qualitative nature of preceding shocks (positive vs negative). A potential source of asymmetry across frequencies may be the different investment horizons on which traders operate. Denoting by g any given connectedness index, the spillover asymmetry measure (SAM) at frequency ω with respect to the sign of a preceding shock may be written as

$$SAM_g(\omega) = S_g^+(\omega) - S_g^-(\omega). \quad (10)$$

Similarly, the SAM for frequencies κ and λ and for the same index g may be written as

$$SAM_g(\omega_{\kappa,\lambda}) = S_g(\omega_{\kappa}) - S_g(\omega_{\lambda}). \quad (11)$$

The statistical significance of the SAMs can be assessed by employing appropriate bootstrap techniques.

3. The data, the empirical models and the results

3.1. The data and the empirical models

The data for the empirical analysis have been obtained from the Bitstamp exchange (platform).⁶ Although more than 200 platforms for cryptocurrency trading exist, the Bitstamp exchange has been selected for two reasons: (a) it is recognized as a leader in price setting (e.g. Giudici and Abu-Hashish, 2019) and (b) it provides both daily and hourly data. The availability of hourly data is very important for the objective of this paper because it makes feasible the assessment of asymmetric connectedness with regard to the qualitative nature price shocks (details are provided later on in this section). The data are prices (in USD) and trading volumes of Bitcoin (BTC), Ethereum (ETH), Ripple (XRP), and Litecoin (LTC) and they refer to the period 1/1/2018 to 3/31/2020. BTC, ETH, XRP, and LTC were among the cryptocurrencies considered in the study of Bouri et al. (2019a) as well. Figs. A.1–A.4 (in the Appendix) present the evolution of the natural logarithms of price and volume for BTC, ETH, XRP, and LTC, respectively, for the period under study. The prices for BTC, ETH, and LTC have followed similar patterns (i.e. a strong downward trend until the end of 2018, a strong upward trend in the first six months of 2019, and a generally downwards trend since then); the price of XRP, however, has exhibited generally a pronounced negative trend. All four cryptocurrencies have shown a clear downward trend in volume over the first six months of 2018 followed by a full or a partial recovery thereafter. For BTC, ETH, and LTC there have been periods in which price and volume moved in the same direction and periods in which they moved in the opposite direction. This appears to be consistent with Blume's et al. (1994) theoretical result that the contemporaneous relationship between returns and volume in asset markets is V-shaped

⁶ <http://www.cryptodatadownload.com/data/bitstamp/>.

(high volume is consistent with both strong positive and strong negative returns). For XRP, however, price and volume have (in most periods) moved in the opposite direction.

In the empirical studies on the relationship between returns (or volatility) and trading activity, a number of alternative specifications have been adopted for the volume variable. Karpoff (1988) and Bouri et al. (2019a), among others, relied on volume changes. Andersen (1996), and Kumar (2017), among others, employed a measure of “detrended” volume. In Kumar (2017) the “detrended” volume was the residuals from the parametric regression of volume on a quadratic trend; in Andersen (1996), it was the ratio of actual trading volume and its expected (“normal”) value, obtained as a centred rolling two-year mean; in Fousekis and Tzaferi (2020) it was the ratio of the actual volume and its fitted value from a parametric regression model. Batten et al. (2019) utilized the notion of “surprise volume” (i.e. the unexpected above-average trading activity) as a proxy for private information flow. Foster (1995) and Andersen (1996) observed that the use of a ratio is preferable since it avoids the negative volume figures associated with the alternative choices. Here, following Fousekis and Tzaferi (2020), the “detrended” volume has been computed as the ratio of the actual volume and the fitted values from the regression of market activity on a quadratic trend.

The hourly (intraday) returns have been employed to separate the daily returns into positive and negative ones. The returns of cryptocurrency i at day t may be expressed as

$$r_{i,t} = \sum_{h=2}^{24} \ln(p_{i,h,t} / p_{i,h-1,t}) = \sum_{h=2}^{24} r_{i,h,t}, \quad (12)$$

where $p_{i,h,t}$ is the price of cryptocurrency i at hour h and at day t . $r_{i,t}$ in (12) may, in turn, be analysed into its positive and negative components as⁷

$$r_{i,t} = \sum_{h=2}^{24} I(r_{i,h,t} \geq 0) r_{i,h,t} + \sum_{h=2}^{24} I(r_{i,h,t} < 0) r_{i,h,t} = r_{i,t}^+ + r_{i,t}^- \quad (13)$$

(where $r_{i,t}^+$ and $r_{i,t}^-$ are the positive and the negative daily returns).

Tables A.1–A.4 (in the Appendix) present descriptive statistics and tests on the distributions of the variables used in the empirical analysis (i.e. “detrended volume”, $r_{i,t}^+$, and $r_{i,t}^-$) of BTC, ETH, XRP, and LTC, respectively. All twelve distributions are leptokurtic (suggesting tails heavier relative to the normal). The distributions of positive returns and “detrended” volume are positively skewed (pointing to very few large observations at the upper tail), while those of negative returns are negatively skewed. In all cases, the null of normality is rejected at any reasonable level of significance.

3.2. The empirical results

Two VAR models have been estimated for each cryptocurrency. One involves the “detrended” volume and the positive returns series, whereas the other involves the “detrended” volume and the negative returns

⁷ The decomposition of daily returns into two components of opposite sign has been conducted here along the lines of the decomposition of the daily variance of returns into positive and negative semi-variances (Barunik et al., 2016).

⁸ Note that in frequency connectedness analysis H serves only as an approximation factor and it has no interpretation as in the time domain. Barunik and Krehlik (2018) suggest setting H sufficiently high to obtain a better approximation at the lower frequencies. High values for H have been employed by Barunik and Krehlik (2018), Barunik and Kocenda (2019), Lovza and Perez-Laborda (2020). Experimentation with shorter horizons (100 and 200) yielded qualitatively very similar results. Gillaizeau et al. (2019) also reported that the outcomes of frequency connectedness analysis tend to be robust to the choice of H .

Table 1

Frequency connectedness (band [1, 5]).

	(1) Positive returns	(2) Negative Returns	(1)–(2) (asymmetry with respect to the sign of price shocks)
(a) BTC			
Total	8.31 (<0.01)	11.29 (<0.01)	–2.98 (0.07)
spillover			
To:			
Returns	3.94 (<0.01)	5.98 (<0.01)	–2.04 (0.02)
Volume	4.37 (<0.01)	5.31 (<0.01)	–0.94 (0.30)
Net	–0.43 (0.43)	0.67 (0.03)	
(b) ETH			
Total	11.31 (<0.01)	14.07 (<0.01)	–2.76 (0.22)
spillover			
To:			
Returns	5.34 (<0.01)	6.80 (<0.01)	–1.46 (0.20)
Volume	5.97 (<0.01)	7.27 (<0.01)	–1.30 (0.30)
Net	–0.63 (0.25)	–0.47 (0.34)	
(c) XRP			
Total	13.22 (<0.01)	11.20 (<0.01)	2.01 (0.07)
spillover			
To:			
Returns	6.25 (<0.01)	5.69 (<0.01)	0.56 (0.35)
Volume	6.97 (<0.01)	5.51 (<0.01)	1.46 (0.04)
Net	–0.72 (0.21)	0.18 (0.69)	
(d) LTC			
Total	10.25 (<0.01)	10.97 (<0.01)	–0.72 (0.67)
spillover			
To:			
Returns	4.89 (<0.01)	5.31 (<0.01)	–0.42 (0.60)
Volume	5.37 (<0.01)	5.66 (<0.01)	–0.29 (0.75)
Net	–0.48 (0.33)	–0.35 (0.25)	

Note: p -values in parentheses. They are obtained as in Patton (2013) using bootstrap with 5000 replications and a Wald-type test which, under the null hypothesis, follows the χ^2 distribution with 1 degree of freedom. The relevant test statistic is $\Phi = (RA)'(RV_{\Lambda}R')^{-1}(RA)$ where R is the restrictions matrix, Λ is the vector of estimates, and V_{Λ} is their bootstrap variance-covariance matrix.

series. The lag order of the VAR models has been determined using the Schwarz criterion and H has been set equal to 300.⁸ Frequency spillovers have been assessed at the bands [1, 5], and (5, 300] days.⁹ The first band corresponds to the small-scale frequencies, while the second to the medium- and the large-scale ones. The estimation of the VAR models and the calculation of spillovers have been carried out using the R package frequencyConnectedness (Krehlik, 2020).

Table 1 shows the estimates of frequency connectedness (which as mentioned in section 2 constitutes a complete measure of the temporal link between two stochastic processes) along with tests for symmetry with respect to the sign of price shocks at the band [1, 5]. Under positive returns, the total spillover index ranges from 8.3 for BTC to 13.2 for XRP; under negative returns, it ranges from 11 for LTC to 14.1 for ETH. The total spillover index is strongly statistically significant everywhere.

All directional spillovers are also statistically significant at any reasonable level pointing, thus, to the presence of bi-directional causality in the volume-returns relationship. From Table 1 follows that shocks in returns (regardless of sign) are likely to explain from 4 to 6.8 per cent of the volume variability whereas shocks to volume are likely to explain from 4.4 to 7.3 of the variability of returns. This finding is in line with the empirical results from several earlier studies in traditional asset markets (e.g. Hiemstra and Jones, 1994; Fujihara and Mougue, 1997; Gebka and Wohar, 2013; He et al., 2014).

⁹ A more detailed decomposition involving three frequency bands has been also conducted. It did not, however, provide any additional insights about the relationship under study.

The bi-directional causality suggests that technical analysis utilizing information of past volume may attain abnormal profits and that past returns may be employed to forecast trading activity and liquidity in cryptocurrency markets as well as to design efficient trading strategies. In their seminal paper, [Blume et al. \(1994\)](#) offered a theoretical justification for the existence of causal order from past trading activity to current returns. In particular, they showed that the level of volume reflects the precision (quality) of information that cannot be deduced from the price statistic alone. For the existence of causal order from past returns to current volume, several alternative explanations have been advanced in the relevant literature including the informativeness of past returns for rational uninformed investors, the liquidity effect, the momentum investment, the over-confidence and the disposition effects (e.g. [Griffin et al., 2007](#))¹⁰. There is some empirical evidence that certain among these effects (factors) may be active in cryptocurrency markets. Specifically, [Park and Chai \(2020\)](#) found that it is the investment sentiment that drives the transaction behaviours of uninformed traders; [Jalal et al. \(2020\)](#) attributed high volumes to overexcitement among investors, whereas [Baur and Dimpfl \(2018\)](#) suggested that the asymmetric impact of positive and negative price changes on price volatility is consistent with the fear of missing out (FOMO) of uninformed traders.

[Gennotte and Leland \(1990\)](#) advanced the misinterpreted-trade hypothesis to explain market crashes. According to it, large negative returns might be the outcome of temporary market illiquidity (traders, occasionally, misinterpret non-informational liquidity trading as being informational; then, even a small amount of uninformed trading may lead to a large price decline). An implication of the misinterpreted-trade hypothesis is that spillovers from volume to returns might be asymmetric (large price increases tend to be accompanied with large trading activity, while large price drops might occur without substantial trading volume). The relevant empirical evidence is far from conclusive. [Marsh and Wagner \(2004\)](#), applying extreme value models to seven stock markets, found that spillovers from volume to returns were asymmetric in the Netherlands, Hong Kong, the UK, and Japan but symmetric in the US, Germany, and France. [Balduzzi et al. \(1996\)](#), using simple linear regressions under crashes and non-crashes, concluded that the misinterpreted-trade hypothesis explained stock market crashes in the US. [Fousekis and Tzaferi \(2020\)](#), applying non-parametric regressions to five energy futures markets in the US, reported that large positive returns occurred together with large levels of trading as often as did large negative returns. The empirical results of [Balcilar et al. \(2017\)](#) and [Bouri et al. \(2019a\)](#) from cryptocurrency markets (discussed in Section 1) are certainly not consistent with the misinterpreted-trade hypothesis.

In this study, the evidence of asymmetric spillovers with respect to the sign of price shocks is both limited and mixed. For BTC, the total and the directional spillover from volume to returns are stronger under negative returns at the 10 and the 5 per cent level of significance, respectively. For the other three cryptocurrencies, the sign of price shocks appears to have no influence whatsoever on the magnitude of spillovers from volume to returns. Our empirical results, therefore, are in line with those of [Fousekis and Tzaferi \(2020\)](#) for the US energy futures markets, of [Balcilar et al. \(2017\)](#) and [Bouri et al. \(2019a\)](#) for cryptocurrency markets, and (to a lesser extent) with those of [Marsh and Wagner \(2004\)](#) for stock markets. Differences in empirical findings may be attributed to differences in

¹⁰ A brief summary of these explanations follows. Rational uninformed investors may find it optimal to adopt trend-following trading strategies if they feel that past prices contain signals about the private information of insiders or about state variables that determine expected returns (rational uninformed investors); Liquidity providers may become more risk-averse as prices decrease. Poor returns reduce volume because the cost of trading rises (liquidity effect); Momentum (or positive feedback) traders are likely to be more active following absolute price moves (momentum trading); Trading should increase with positive returns if such returns build confidence (over-confidence effect); Volume follows returns because investors are eager to lock in gains after price increases but reluctant to trade after price decreases (disposition effect).

Table 2

Frequency connectedness (band (5, 300)).

	(1) Positive returns	(2) Negative returns	(1)–(2) (asymmetry with respect to the sign of price shocks)
(a) BTC			
Total	19.67	19.69 (<0.01)	–0.02 (0.99)
spillover	(<0.01)		
To:			
Returns	6.26 (<0.01)	9.21 (<0.01)	–2.95 (0.24)
Volume	13.41	10.48 (<0.01)	2.93 (0.09)
	(<0.01)		
Net	–7.15 (0.02)	–1.27 (0.41)	
spillover			
(b) ETH			
Total	19.90	19.99 (<0.01)	–0.09 (0.97)
spillover	(<0.01)		
To:			
Returns	5.16 (<0.01)	9.21 (<0.01)	–4.05 (0.13)
Volume	14.75	10.79 (<0.01)	3.96 (0.02)
	(<0.01)		
Net	–9.59	–1.58 (0.39)	
spillover	(<0.01)		
(c) XRP			
Total	17.55	14.73 (<0.01)	2.82 (0.15)
spillover	(<0.01)		
To:			
Returns	7.62 (<0.01)	6.31 (<0.01)	1.31 (0.45)
Volume	9.93 (<0.01)	8.42 (<0.01)	1.51 (0.19)
Net	–2.31 (0.10)	–2.11 (0.22)	
spillover			
(d) LTC			
Total	15.97	14.81 (<0.01)	1.16 (0.61)
spillover	(<0.01)		
To:			
Returns	5.23 (<0.01)	5.60 (<0.01)	–0.36 (0.86)
Volume	10.74	9.21 (<0.01)	1.53 (0.23)
	(<0.01)		
Net	–5.51 (0.01)	–3.61	
spillover		(<0.01)	

Note: *p*-values in parentheses. They are obtained as in [Patton \(2013\)](#) using bootstrap with 5000 replications and a Wald-type test which, under the null hypothesis, follows the χ^2 distribution with 1 degree of freedom. The relevant test statistic is $\Phi = (RA)'(RV_A R')^{-1}(RA)$ where *R* is the restrictions matrix, *A* is the vector of estimates, and *V_A* is their bootstrap variance-covariance matrix.

the characteristics of the studied markets (stocks, commodities, cryptocurrencies) and the methodological approaches employed. For XRP, the total and the directional spillover from returns to volume are stronger under positive returns at the 10 and the 5 per cent level of significance, respectively. The net spillovers are, generally, not significant indicating that, at the high frequencies, the contributions of returns and volume to the total spillover are about the same (exception is BTC under negative returns only).

Table 2 presents estimates of frequency connectedness along with tests for symmetry with respect to the sign of price shocks at the band (5, 300]. The total spillovers measure ranges from 14.7 to 20. In all cases, the directional spillovers from returns to volume are higher than those in the opposite direction. Under positive (negative) returns, three (one) net spillovers are statistically significant at the 5 per cent level. These results suggest that, at the medium and the low frequencies, shocks in returns are more important for the connectivity of the volume-returns stochastic system relative to shocks in volume; as a consequence, not considering the causal order from returns to volume and the existence of feedback is a potentially serious omission. The evidence of asymmetry with respect to the sign of price shocks at the frequency band (5, 300] is even weaker relative to that for the high frequencies¹¹; in just two cases (directional

¹¹ According to [Foster \(1995\)](#) and to [Fousekis and Tzaferi \(2020\)](#) a symmetric impact of positive and negative returns on volume is also a form of market efficiency.

spillovers from returns to volume for ETH and BTC) the differences are statistically significant at the conventional levels, indicating that positive returns have a higher impact relative to negative ones.

Figs. A.5–A.6 (in the Appendix) present rolling window estimates for total spillover, spillover to volume, and spillover to returns by the sign of the preceding price shock for the frequency bands [1,5] and (5, 300], respectively. As in Barunik and Krehlik (2018), and Barunik and Kocenda (2019) the window length has been set equal to 300.

Table 3 presents the symmetry test with respect to frequency by the nature of price shocks. The spillovers (either total or directional) are all higher at the medium and the low frequencies. With regard to the total measure of connectedness, all but one (that for XRP under positive returns) differences are statistically significant at the conventional levels. With regard to the directional measures, the symmetry of spillovers from returns (irrespective of the sign) to volume is rejected everywhere and, in most cases, very strongly. The symmetry of spillovers, however, from volume to returns is rejected only for BTC under negative returns. As noted by Balke and Wohar (2002) and Barunik and Krehlik (2018) connectivity at high frequencies occurs when markets process information very rapidly and shocks affect traders' short-run behaviour, whereas connectivity at low frequencies results from fundamental changes in traders' expectations. The findings here suggest that shocks to the system may induce fundamental changes in expectations. These changes, however, are more likely to be associated with shocks to prices than with shocks to volume. Moreover, the differences in spillovers (total and directional) between the larger- and smaller-scale frequencies are invariably higher under positive returns indicating that, for investors with longer horizons, a good performance of the market in the past influences their trading activity more than a bad one. Changes in volume, on the other hand, appear not to have any differential impact related to the investment horizon of the participants in cryptocurrency markets.

Table 4 presents the power spectral density of the system returns-volume for the four cryptocurrencies considered. In all cases, the larger-scale frequencies account for well above 50 per cent of the strength of variation (energy of the system). Granger (1966) argued that the typical spectral of economic variables concentrates spectral mass at lower frequencies, declining smoothly at higher ones. Medel (2014)

Table 3
Frequency connectedness. Tests of symmetry with respect to frequency band.

	(5, 300] – [1,5]	
	Positive returns	Negative returns
(a) BTC		
Total spillover	11.36 (<0.01)	8.40 (<0.01)
To:		
Returns	2.32 (0.12)	3.23 (0.04)
Volume	9.04 (<0.01)	5.17 (<0.01)
(b) ETH		
Total spillover	8.59 (<0.01)	5.92 (0.04)
To:		
Returns	–0.18 (0.91)	2.41 (0.25)
Volume	8.77 (<0.01)	3.51 (0.02)
(c) XRP		
Total spillover	4.33 (0.15)	3.53 (0.09)
To:		
Returns	1.37 (0.44)	0.62 (0.66)
Volume	2.96 (0.06)	2.91 (0.04)
(d) LTC		
Total spillover	5.71 (0.01)	3.83 (0.10)
To:		
Returns	0.35 (0.80)	0.29 (0.84)
Volume	5.37 (<0.01)	3.55 (<0.01)

Note: *p*-values in parentheses. They are obtained as in Patton (2013) using bootstrap with 5000 replications and a Wald-type test which, under the null hypothesis, follows the χ^2 distribution with 1 degree of freedom. The relevant test statistic is $\Phi = (RA)'(RV_{\Lambda}R')^{-1}(RA)$ where *R* is the restrictions matrix, Λ is the vector of estimates, and V_{Λ} is their bootstrap variance-covariance matrix.

Table 4

Power spectral density for the system returns - volume Positive returns Negative returns.

Frequency band	[1, 5]	(5, 300]	[1, 5]	(5, 300]
BTC	0.357	0.642	0.369	0.631
ETH	0.404	0.595	0.417	0.583
XRP	0.421	0.579	0.403	0.597
LTC	0.422	0.577	0.415	0.585

investigated the spectral shape of a very large number of variables from the real economy, banking, and finance. He found that while for the real economy variables (e.g. industrial production index, civilian unemployment index) almost all energy was concentrated at the large-scale frequencies, for finance variables (e.g. S&P 500 stock price index, VIX) small-scale frequencies contributed a sizable part of total energy. Here, the contribution of higher frequencies ranges from 36 to 42 per cent. Therefore, although the longer-run component dominates, the shorter-run one (noise) is also important for the system dynamics; as far as the spectral pattern is concerned, returns and volume in cryptocurrencies behave like financial variables. A possible (technical) explanation of this finding is that returns and “detrended” volume are time series with a relatively low level of persistence.

Tables A.5–A.7 (in the Appendix) present the results from the within frequency causation analysis. Note that the spillovers, in this case, are just the spillovers from the frequency connectedness analysis divided by the respective shares of the smaller- and the larger-scale frequencies. Depending on the frequency band and the cryptocurrency considered, the total and the directional spillovers are from 60 to 185 per cent higher than the corresponding in Tables 1–2. Relative to the results of frequency connectedness analysis, returns is still a net contributor of spillovers and volume is a net receiver (especially at the lower frequencies). Also, positive and negative returns appear to have similar impacts. The evidence of asymmetry with respect to frequency band, however, is somehow weaker but it is still driven mainly by the positive shocks.

4. Conclusions

The temporal relationship between returns and volume is important for assessing informational efficiency, forecasting trading activity, and devising investment strategies in asset markets. This study has investigated the lag-lead association of returns and volume in major cryptocurrency markets using data over 2018 to 2020 and frequency connectedness analysis. Special care has been taken to identify and quantify potential asymmetries with regard to traders' investment horizon, the qualitative nature of price shocks (positive vs negative), and the source of spillovers (volume vs returns).

The empirical results suggest:

- Causality is bi-directional (the system returns-volume is a feedback one). The existence of causality from volume to returns implies that volume has an information content which, when exploited, may help traders to achieve higher profit. The existence of causality from returns to volume suggests that the performance of a cryptocurrency market in the past may impact on future volume and investment strategies. The causal order from returns to volume is stronger than that in the opposite direction but only for larger-scale frequencies; for higher frequencies, the directional spillovers are, generally, of very similar magnitude. The finding that there is causal order from returns to volume is consistent with theoretical explanations and recent empirical results that highlight the importance of sentiment, trend-following (and even herding) behaviour of uninformed traders for the level of transactions in cryptocurrency markets. These influences may create a high-risk environment. It appears, therefore, that there is scope for mitigating information asymmetry through the establishment of

appropriate systems (such as the Corporate Disclosure, the Regulation Fair Disclosure, and the Financial Information Disclosure that have long been available for the traditional asset markets).

- (b) For smaller-scale frequencies, there is some evidence of asymmetry with respect to the sign of the preceding price shocks. It is not, however, consistent across cryptocurrencies; for BTC, negative returns give rise to larger spillovers while the opposite appears to be the case for XRP.
- (c) Although the larger-scale frequency components dominate the strength of the system's variability, the contribution of the smaller-scale ones is very far from negligible. The spectral pattern is akin to that of financial variables with a not very long memory.
- (d) For frequency connectedness (which is the most complete of the two measures of connectivity considered here), there is considerable evidence of asymmetry with regard to the scale of frequencies. Total and directional spillovers from returns to volume are much stronger at the lower frequency band while those from volume to returns are of similar magnitude in the two time-scales. Fundamental changes of expectations, therefore, are more likely to occur through returns than through volume changes. Moreover, there is some evidence that between positive and negative returns the former may have a stronger impact on trading activity in the longer-run.
- (e) The evidence of asymmetric spillovers from volume to returns under price increases and decreases is limited and mixed. The empirical results, therefore, do not appear to support the misinterpreted-trade hypothesis for market crashes.

There is a number of potential avenues for future research. One may involve the use of frequency connectedness analysis to investigate the lag-lead relationship between volume and returns or volatility in other asset markets. Another may compare the empirical results on the returns-volume link in cryptocurrency markets from alternative parametric and non-parametric approaches. Given that the temporal relation between returns and trading activity is important for market efficiency and for investment strategies further research on this elaborate topic is certainly warranted.

Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.econmod.2020.11.013>.

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